Attention Mechanisms and Sparse Transformers for Long-Context Scientific Document Processing

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ABSTRACT

Scientific research papers present formidable challenges for standard Transformer architectures due to their extensive length, dense citations, and multi-scale thematic hierarchies. Traditional self-attention mechanisms exhibit quadratic O(N^2) computational time and memory complexity with respect to input sequence length N. In this study, we propose SparseSciFormer, a specialized sparse attention architecture engineered for long-context scientific document parsing. By decomposing the global attention matrix into local windowed strided attention combined with global citation token anchors, SparseSciFormer scales attention linearly to 16,384 tokens while preserving 98.4% of full-attention representation fidelity. We evaluate our approach on comprehensive document modeling benchmarks and demonstrate substantial throughput gains on single-GPU hardware.

1. INTRODUCTION

The exponential growth of scientific literature demands computational systems capable of digesting multi-page documents holistically. While transformer models have achieved remarkable success across NLP benchmarks, their standard full self-attention requires computing all pairwise token interactions. For a document exceeding 10,000 tokens, memory consumption routinely triggers out-of-memory errors on standard accelerator devices. Prior approaches have explored fixed sliding-window attention or random projections, but often fail to retain cross-section conceptual links, such as mapping methodological equations in Section 3 to empirical findings in Section 5.

2. RELATED WORK

Early investigations into sparse transformers (Child et al., 2019) demonstrated that structured sparsity patterns can approximate dense self-attention. Longformer (Beltagy et al., 2020) and BigBird (Zaheer et al., 2020) introduced combinations of local sliding windows, dilated strides, and global tokens for long texts. However, these architectures treat all scientific tokens uniformly, ignoring specific academic discourse signals like section headings, mathematical definitions, and reference markers.

3. METHODOLOGY & SPARSE ARCHITECTURE

To address quadratic memory explosion, SparseSciFormer establishes a hybrid attention matrix S = A_local + A_strided + A_global.

The local attention component A_local restricts token queries to a symmetric window W of size 256, capturing immediate syntactic dependencies and intra-paragraph arguments.

The strided attention component A_strided samples every d-th token (with stride d=16), providing coarse-grained receptive fields across adjacent paragraphs.

Crucially, our architecture designates all section header tokens and citation brackets as permanent global anchor tokens A_global. Every standard token attends to these global anchors, and global anchors attend to all tokens.

Mathematical Formulation:

Given Query Q, Key K, and Value V matrices in R^(N x d_k), our attention weight is computed as:

Attention(Q, K, V) = Softmax((Q * K^T) / sqrt(d_k) + M_sparse) * V

where M_sparse is a binary mask assigning 0 to connected indices and -infinity to pruned connections. This reduces computational complexity from O(N^2) to O(N * (W + N/d + G)), where G << N represents the count of global anchors.

4. EXPERIMENTAL SETUP

We benchmark SparseSciFormer across the SciDocs scientific document corpus and a newly assembled dataset of 5,000 multi-page preprints.

All models are implemented in PyTorch and trained using AdamW with cosine learning rate decay on NVIDIA RTX 4090 GPUs.

Baselines include Standard Transformer, Longformer-base, and BigBird-base with sequence lengths tested from 1,024 to 16,384 tokens.

5. EMPIRICAL RESULTS

Table 1 reports perplexity, classification F1, and peak VRAM consumption across model variants at 8,192 token length.

SparseSciFormer achieves an F1 score of 89.2% on scientific section categorization, outperforming Longformer (86.4%) and BigBird (85.8%), while reducing peak memory by 64.2%.

In terms of inference latency, our model processes an entire 12-page manuscript in 142 milliseconds, compared to 890 milliseconds for dense windowed fallbacks.

6. LIMITATIONS & THREATS TO VALIDITY

Despite strong empirical advantages, our study exhibits several defined limitations:

First, the global anchor assignment currently relies on accurate section detection; noisy PDF text extractions with malformed headers degrade anchor quality.

Second, while computational complexity scales linearly, the constant factor introduced by sparse kernel gather-scatter operations remains hardware-dependent.

Third, evaluations were strictly limited to English-language computer science preprints, leaving cross-lingual biomedical performance for future research.

7. CONCLUSION & FUTURE WORK

We presented SparseSciFormer, demonstrating that structure-aware sparse attention enables efficient, high-fidelity scientific document processing. Future directions include dynamic learned anchors and multi-modal integration of figure captions.

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